# CSC311: Introduction to Machine Learning

## Project Report

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# 1 Data Exploration

## Dataset Overview

- **Dataset Composition**: The dataset contains 870 survey responses collected from 290 distinct students, with each respondent evaluating all three sports (Soccer, Baseball, and Basketball). In this dataset, there are 14 columns representing the survey questions and responses.
- **Class Balance**: The target labels are perfectly balanced (290 observations per sport overall; exactly 203 per sport in the 70% training split). No class rebalancing or oversampling is needed.

## Data Splitting

- **Respondent Isolation (`id`)**: The respondent identifier (`id`) was used to perform grouped partitioning. All responses from a given student were kept within the same dataset split to prevent data leakage arising from respondent-specific vocabulary, preferences, or writing styles.
- **Train-Test-Validation Split**: The 290 respondent IDs are partitioned into:
  - **Training split (70%)**: 203 respondents (609 observations) used for exploratory analysis, distribution inspection, and feature engineering.
  - **Validation split (15%)**: 43 respondents (129 observations) reserved strictly for hyperparameter tuning.
  - **Internal test split (15%)**: 44 respondents (132 observations) held out for final evaluation of the best model.

  Following partitioning, the validation and test sets were excluded from all exploratory analysis. Feature engineering, imputation parameters, standardization statistics, and text vocabularies were learned exclusively from the training set.

## Features Types

**Table 1:** Feature categories in the dataset.

| Feature Category | Variables |
|---|---|
| Numerical | difficulty_rating |
| Likert-scale | strategy_importance, injury_risk_comfort, individual_brilliance |
| Multi-select categorical | play_location, play_season, associated_music |
| Free-text | associated_foods_drinks, associated_color, associated_song, emotional_response |
| Semi-structured text | acceptable_price |
| Target | label |

## Features Distributions

The distribution of the difficulty ratings follow roughly a normal distribution but appear skewed for baseball and basketball. The Likert-scale features are mostly unimodal, with a slight skew towards higher values for strategy importance and injury risk comfort. Multi-select categorical features show a wide variety of responses, with some locations (e.g. outdoor spaces), music genres (e.g. hip-hop or rap), seasons (e.g. summer) being more popular than others, namely winter and classical/jazz. Free-text features exhibit high variability in length and content, with some respondents providing detailed descriptions while others leave fields blank. Ticket price willingness exhibits extreme positive skewness (+10.5), spanning from $0 to $10,000.

## Missing Values, Outliers, and Inconsistencies

Blank responses are rare overall (∼1.70%), with the song feature exhibiting the highest vacancy (∼90 entries missing). Rather than discarding sparse text rows, a binary indicator (`is_missing`) is appended to each text feature, allowing models to leverage deliberate omission as a predictive signal (e.g., respondents unable to name a song for less mainstream sports). Imputation is also used to handle missing values in numerical and Likert-scale features, with mean imputation for difficulty rating and neutral-value imputation for Likert features.

Outliers are present in the acceptable price feature, with some respondents reporting extreme willingness to pay. A $\log(1 + x)$ transformation is applied to reduce skewness and mitigate outlier influence. Inconsistencies are also present in this feature with some respondents reporting prices with dollar signs, commas, etc. These are stripped retaining only the numeric value during preprocessing.

## Feature Preprocessing

**Table 2:** Feature preprocessing summary.

| Feature Group | Preprocessing | Rationale |
|---|---|---|
| Respondent ID | Excluded from predictors; used for grouped partitioning | Prevents overfitting (IDs are not generalizable). |
| Difficulty Rating | Mean imputation ($\mu = 5.48$); z-score standardization | Removes scale disparity and handles missing values. |
| Likert Features | Monotonic mapping (1–5); neutral imputation (3) | Preserves ordinal structure while avoiding unnecessary dimensionality. |
| Multi-select Features | Lowercasing; delimiter splitting; multi-hot encoding | Represents multiple simultaneous selections without imposing ordering. |
| Text Features | Lowercasing; punctuation removal; TF–IDF vectorization | Downweights common words and emphasizes discriminative sport-specific terms. |
| Acceptable Price | Regex removal of "$" and commas; conversion of "thousand" expressions; median imputation; $\log(1 + x)$ transformation | Produces numeric values and reduces extreme positive skewness. |
| Target Label | Map Soccer → 0, Baseball → 1, Basketball → 2 | Creates machine-readable class labels. |

# 2 Methodology

Three model families will be evaluated: Multinomial Logistic Regression, Decision Trees, and Random Forests.

- **Multinomial Logistic Regression**: A linear classification model that models class probabilities. This model typically performs well on high-dimensional sparse feature spaces produced by TF–IDF representations. L1 regularization can be used to select irrelevant features and L2 regularization can be used to prevent overfitting.
- **Decision Tree**: A non-linear model capable of capturing complex feature interactions while remaining highly interpretable. Decision trees make a good baseline for small datasets as they can capture non-linear relationships without requiring extensive hyperparameter tuning and can handle both categorical (e.g. play_season) and continuous (e.g. acceptable_price) features without the need for scaling or transformation. They do risk overfitting, however, if they are not pruned properly.
- **Random Forest**: An ensemble of decision trees that reduces overfitting and variance while retaining the ability to model non-linear relationships. Bagging can be used to create this random forest, and a random subset of features per tree will be considered to reduce correlation between trees. Random forests are often more accurate than single decision trees, but they are less interpretable and require more computational resources.

**Implementation Procedure**

1. Train each model using the training set only.
2. Perform hyperparameter tuning using the validation set:
    - Logistic Regression: regularization type (L1, L2) and different values of $\lambda$ will experimented with.
    - Decision Tree: maximum depth, minimum samples per split, minimum samples per leaf, and split criterion (Information Gain or Gini).
    - Random Forest: number of trees, maximum depth, minimum samples per split, minimum samples per leaf, and maximum features per split.
3. Evaluate each configuration on the validation set and choose the best model. Metrics to be reported include Accuracy, Precision, Recall, Macro F1-score, and the Confusion Matrix.
4. Select the model and hyperparameter configuration achieving the highest validation Macro F1-score.
5. Retrain the selected model using the combined training and validation data.
6. Evaluate the final model once on the held-out test set and report the values for the evaluated metrics.